

FDSN Reserved Headers

The “**FDSN**” key at the root of the [Extra Headers](#) are reserved for definition by the FDSN. See [A: Extra header examples](#) for an example of FDSN extra headers.

The FDSN reserved headers are defined and documented in [JSON Schema](#).

Download the [FDSN Extra Header schema v1.0](#).

The documentation and schema of these headers may be browsed here:

Expand all

Collapse all

Type: object

Extra headers in miniSEED 3 data format

FDSN

root → [FDSN](#)

Type: object

Reserved extra headers defined by the FDSN

No Additional Properties

Time

root → [FDSN](#) → [Time](#)

Type: object

Headers related to data timing and recording system clock

No Additional Properties

Quality

root → [FDSN](#) → [Time](#) → [Quality](#)

Type: integer

DEPRECATED Timing quality. A vendor specific value from 0 to 100% of maximum accuracy. [same as SEED 2.4 Blockette 1001, field 3] It is recommended to use MaxEstimatedError instead.

Correction

root → [FDSN](#) → [Time](#) → [Correction](#)

Type: number

Time correction applied to record start time in seconds. [same as SEED 2.4 FSDH, field 16]

MaxEstimatedError

root → [FDSN](#) → [Time](#) → [MaxEstimatedError](#)

Type: number

Maximum estimated timing error in seconds.

LeapSecond

root → [FDSN](#) → [Time](#) → [LeapSecond](#)

Type: integer

If present, one or more leap seconds occuring during this record. The value specifies the number of leap seconds and direction. For example, use 1 to specify a single positive leap second and -1 to specify a single negative leap second. [incorporates SEED 2.4 FSDH, field 12, bits 4 and 5]

Exception

root → [FDSN](#) → [Time](#) → [Exception](#)

Type: array of object

List of timing exceptions

No Additional Items

Each item of this array must be:

When not present, the boolean values in the FDSN reserved headers should be considered to be *false* unless otherwise documented. Such values do not need to be included when the value is *false*.

Type: object

No Additional Properties

Previous

Next

Time

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root → [FDSN](#) → [Time](#) → [Exception](#) → [Exception items](#) → [Time](#)

Type: string

Format: date-time

Time of timing exeption. [same as SEED 2.4 Blockette 500, field 4]

VCOCorrection

root → [FDSN](#) → [Time](#) → [Exception](#) → [Exception items](#) → [VCOCorrection](#)

Type: number

VCO correction, a percentage from 0 to 100% of VCO control value, where 0 is the slowest and 100 is the fastest. [same as SEED 204 Blockette 500, field 3]

ReceptionQuality

root → [FDSN](#) → [Time](#) → [Exception](#) → [Exception items](#) → [ReceptionQuality](#)

Type: integer

Reception quality as a percent of maximum clock accuracy based only on information from the clock. [same as SEED 2.4 Blockette 500, field 6]

Count

root → [FDSN](#) → [Time](#) → [Exception](#) → [Exception items](#) → [Count](#)

Type: integer

Exception count, with meaning based on type of exception, such as 15 missing time marks. [same as SEED 2.4 Blockette 500, field 7]

Type

root → [FDSN](#) → [Time](#) → [Exception](#) → [Exception items](#) → [Type](#)

Type: string

Description of clock exception, such as MISSING TIMEMARK. [same as SEED 2.4 Blockette 500, field 8]

ClockStatus

root → [FDSN](#) → [Time](#) → [Exception](#) → [Exception items](#) → [ClockStatus](#)

Type: string

Description of clock status, clock specific parameters such as the station for an Omega clock or satellite signal to noise ratios for GPS clocks. [same as SEED 2.4 Blockette 500, field 10]

Event

root → [FDSN](#) → [Event](#)

Type: object

Headers related to event detection and progression

No Additional Properties

Begin

root → [FDSN](#) → [Event](#) → [Begin](#)

Type: boolean

An event starts in this record. [same as SEED 2.4 FSDH, field 12, bit 2]

End

root → [FDSN](#) → [Event](#) → [End](#)

Type: boolean

An in-progress event ends in this record, i.e. the detection algorithm de-triggers. [same as SEED 2.4 FSDH, field 12, bit 3]

InProgress

root → [FDSN](#) → [Event](#) → [InProgress](#)

Type: boolean

An event is in progress. [same as SEED 2.4 FSDH, field 12, bit 6]

Detection

root → [FDSN](#) → [Event](#) → [Detection](#)

Type: array of object

List of event detections

No Additional Items

Each item of this array must be:

root → [FDSN](#) → [Event](#) → [Detection](#) → [Detection items](#)

Type: object

No Additional Properties

Type

root → [FDSN](#) → [Event](#) → [Detection](#) → [Detection](#)

items → Type Type: string Type of event detection, e.g. MURDOCK
SignalAmplitude
root → FDSN → Event → Detection → Detection items → SignalAmplitude Type: number Event amplitude of signal in counts. [same as SEED 2.4 Blockettews 200 and 201, field 3]
SignalPeriod
root → FDSN → Event → Detection → Detection items → SignalPeriod Type: number Event period of signal in seconds. [same as SEED 2.4 Blockettews 200 and 201, field 4]
BackgroundEstimate
root → FDSN → Event → Detection → Detection items → BackgroundEstimate Type: number Event background estimate in counts. [same as SEED 2.4 Blockettews 200 and 201, field 5]
Wave
root → FDSN → Event → Detection → Detection items → Wave Type: string Event detection wave, use values of DILATATION or COMPRESSION if determined. [same as SEED 2.4 Blockettews 200 and 201, field 6, bit 0]
Units
root → FDSN → Event → Detection → Detection items → Units Type: string Units of amplitude band background estimate, use COUNTS for unprocessed signal. [similar to SEED 2.4 Blockettews 200, field 6, bit 1]
OnsetTime
root → FDSN → Event → Detection → Detection items → OnsetTime Type: stringFormat: date-time Event signal onset time. [same as SEED 2.4

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		Blockette 200 and 201, field 8]	
		MEDSNR	
		root → FDSN → Event → Detection → Detection items → MEDSNR Type: array of number Murdock event detection signal-to-noise ratios. [same as SEED 2.4 Blockette 201, field 9] No Additional Items Each item of this array must be: root → FDSN → Event → Detection → Detection items → MEDSNR → MEDSNR items Type: number	
		MEDLookback	
		root → FDSN → Event → Detection → Detection items → MEDLookback Type: integer Murdock event detction lookback value: 0, 1 or 2. [same as SEED 2.4 Blockette 201, field 10]	
		MEDPickAlgorithm	
		root → FDSN → Event → Detection → Detection items → MEDPickAlgorithm Type: integer Murdock event detection pick algorithm: 0 or 1. [same as SEED 2.4 Blockette 201, field 11]	
		Detector	
		root → FDSN → Event → Detection → Detection items → Detector Type: string Name of the event detector. [same as SEED 2.4 Blockette 200, field 9 and Blockette 201, field 12]	
		Calibration	
		root → FDSN → Calibration Type: object Headers related to calibrations No Additional Properties Sequence	latest
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root → [FDSN](#) → [Calibration](#) → [Sequence](#)

Type: array of object

List of calibration sequences

No Additional Items

Each item of this array must be:

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#)

Type: object

List of calibrations

No Additional Properties

Type

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → Type

Type: string

Type of calibration: STEP, SINE, PSEUDORANDOM, GENERIC.

BeginTime

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → BeginTime

Type: string Format: date-time

Calibration begin time. [similar to SEED 2.4 Blockettes 300,310,320,390, field 3]

EndTime

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → EndTime

Type: string Format: date-time

Calibration end time. [same as SEED 2.4 Blockette 395, field 3]

Steps

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → Steps

Type: number

Number of step calibrations in sequence. [same as SEED 2.4 Blockette 300, field 4]

StepFirstPulsePositive

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → StepFirstPulsePositive

Type: boolean

Step calibration, first pulse is positive. [same as SEED 2.4 Blockette 300, field 5, bit 0]

StepAlternateSign

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → [StepAlternateSign](#)

Type: boolean

Step calibration alternate sign. [same as SEED 2.4 Blockettes 300, field 5, bit 1]

Trigger

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → [Trigger](#)

Type: string

Calibration trigger, use AUTOMATIC or MANUAL. [same as SEED 2.4 Blockettes 300,310,320,390, field 5, bit 2]

Continued

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → [Continued](#)

Type: boolean

Calibration is continued from previous records. [same as SEED 2.4 Blockettes 300,310,320,390, field 5, bit 3]

Amplitude

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → [Amplitude](#)

Type: number

Amplitude of calibration signal. For pseudo-random calibrations, this should be the peak-to-peak amplitude of the steps. [same as SEED 2.4 Blockettes 300,310, field 8 and Blockettes 320,390, field 7]

InputUnits

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → [InputUnits](#)

Type: string

Units of calibration input, usually volts or amps. [same as SEED 2.4 Blockette 52, field 9]

AmplitudeRange

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → [AmplitudeRange](#)

Type: string

Amplitude range of calibration, use values of PEAKTOPEAK, ZEROTOPEAK, RMS OR RANDOM. [same as SEED 2.4 Blockette 310, field 5,

bits 4,5,6 and Blockette 320, field 5, bit 4]
Duration
root → FDSN → Calibration → Sequence → Sequence items → Duration Type: number Duration of calibration in seconds. For step calibrations this is the duration of the step. [same as SEED 2.4 Blockettes 300,310,320,390, field 6]
SinePeriod
root → FDSN → Calibration → Sequence → Sequence items → SinePeriod Type: number Period of sine calibrations in seconds. [same as SEED 2.4 Blockette 310, field 7]
StepBetween
root → FDSN → Calibration → Sequence → Sequence items → StepBetween Type: number Interval between times a step calibration is on in seconds. [same as SEED 2.4 Blockette 300, field 7]
InputChannel
root → FDSN → Calibration → Sequence → Sequence items → InputChannel Type: string Channel containing the calibration input. [same as SEED 2.4 Blockettes 300,310, field 9 and Blockettes 320,390, field 8]
ReferenceAmplitude
root → FDSN → Calibration → Sequence → Sequence items → ReferenceAmplitude Type: number Reference amplitude. This is a user-defined value that indicates either the voltage or amperage of the calibration signal when the calibrator is set to 0dB. [same as SEED 2.4 Blockettes 300,310, field 11 and Blockette 320, field 10]
Coupling
root → FDSN → Calibration → Sequence → Sequence items → Coupling Type: string

Coupling of the calibration signal, such as RESISTIVE or CAPACITIVE. [same as SEED 2.4 Blockettes 300,310, field 12 and Blockette 320, field 11]

Rolloff

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → [Rolloff](#)

Type: string

Rolloff characteristics for any filters used on the calibrator, such as 3dB@10Hz. [same as SEED 2.4 Blockettes 300,310, field 13 and Blockette 320, field 12]

Noise

root → [FDSN](#) → [Calibration](#) → [Sequence](#) → [Sequence items](#) → [Noise](#)

Type: string

Noise characteristics for pseudo-random calibrations, such as WHITE or RED. [same as SEED 2.4 Blockette 320, field 13]

Recenter

root → [FDSN](#) → [Recenter](#)

Type: object

Headers related to sensor recentering (mass, gimble, etc.)

No Additional Properties

Sequence

root → [FDSN](#) → [Recenter](#) → [Sequence](#)

Type: array of object

List of recentering sequences

No Additional Items

Each item of this array must be:

root → [FDSN](#) → [Recenter](#) → [Sequence](#) → [Sequence items](#)

Type: object

List of recenterings

No Additional Properties

Type

root → [FDSN](#) → [Recenter](#) → [Sequence](#) → [Sequence items](#) → [Type](#)

Type: string

Type of recenter: Mass, Gimbal, etc. If omitted a mass recenter may be assumed.

BeginTime

root → [FDSN](#) → [Recenter](#) → [Sequence](#) → [Sequence items](#) → [BeginTime](#)
Type: string **Format:** date-time
Recenter begin time.

EndTime

root → [FDSN](#) → [Recenter](#) → [Sequence](#) → [Sequence items](#) → [EndTime](#)
Type: string **Format:** date-time
Estimate of recenter end time.

Trigger

root → [FDSN](#) → [Recenter](#) → [Sequence](#) → [Sequence items](#) → [Trigger](#)
Type: string
Calibration trigger, use AUTOMATIC or MANUAL.

Flags

root → [FDSN](#) → [Flags](#)
Type: object
No Additional Properties

MassPositionOffscale

root → [FDSN](#) → [Flags](#) → [MassPositionOffscale](#)
Type: boolean
Sensor mass position is offscale as defined by the vendor or operator.

AmplifierSaturation

root → [FDSN](#) → [Flags](#) → [AmplifierSaturation](#)
Type: boolean
Amplifier saturation detected. [same as SEED 2.4 FSDH, field 14, bit 0]

DigitizerClipping

root → [FDSN](#) → [Flags](#) → [DigitizerClipping](#)
Type: boolean
Digitizer clipping detected. [same as SEED 2.4 FSDH, field 14, bit 1]

Spikes

root → [FDSN](#) → [Flags](#) → [Spikes](#)

Type: boolean

Spikes detected. [same as SEED 2.4 FSDH, field 14, bit 2]

Glitches

root → [FDSN](#) → [Flags](#) → [Glitches](#)

Type: boolean

Glitches detected. [same as SEED 2.4 FSDH, field 14, bit 3]

FilterCharging

root → [FDSN](#) → [Flags](#) → [FilterCharging](#)

Type: boolean

A digital filter may be charging. [same as SEED 2.4 FDSH, field 14, bit 6]

StationVolumeParityError

root → [FDSN](#) → [Flags](#) → [StationVolumeParityError](#)

Type: boolean

DEPRECATED Station volume parity error possibly present. [same as SEED 2.4 FDSH, field 13, bit 0]

LongRecordRead

root → [FDSN](#) → [Flags](#) → [LongRecordRead](#)

Type: boolean

DEPRECATED Long record read (possibly no problem). [same as SEED 2.4 FDSH, field 13, bit 1]

ShortRecordRead

root → [FDSN](#) → [Flags](#) → [ShortRecordRead](#)

Type: boolean

DEPRECATED Short record read (record padded). [same as SEED 2.4 FDSH, field 13, bit 2]

StartOfTimeSeries

root → [FDSN](#) → [Flags](#) → [StartOfTimeSeries](#)

Type: boolean

DEPRECATED Start of time series. [same as SEED 2.4 FDSH, field 13, bit 3]

EndOfTimeSeries

root → FDSN → Flags → EndOfTimeSeries Type: boolean DEPRECATED End of time series. [same as SEED 2.4 FDSH, field 13, bit 4]
MissingData
root → FDSN → Flags → MissingData Type: boolean DEPRECATED Missing data. [same as SEED 2.4 FDSH, field 14, bit 4]
TelemetrySyncError
root → FDSN → Flags → TelemetrySyncError Type: boolean DEPRECATED Telemetry synchronization error. [same as SEED 2.4 FDSH, field 14, bit 5]

Logger

root → [FDSN](#) → [Logger](#)

Type: object

Description of data logger

No Additional Properties

Model

root → [FDSN](#) → [Logger](#) → [Model](#)

Type: string

Model of equipment

Serial

root → [FDSN](#) → [Logger](#) → [Serial](#)

Type: string

Serial number of equipment

Sensor

root → [FDSN](#) → [Sensor](#)

Type: object

Description of sensor

[Same definition as Logger](#)

Clock

root → [FDSN](#) → [Clock](#)

Type: object

	Description of clock system Same definition as Logger
	ProvenanceURI
	root → FDSN → ProvenanceURI Type: string An identifier for a provenance description
	DataQuality
	root → FDSN → DataQuality Type: string Data quality indicator, use D, R, Q or M. [same as SEED 2.4 FSDH, field 2]
	Sequence
	root → FDSN → Sequence Type: integer Data record sequence number. [same as SEED 2.4 FSDH, field 1]